PART A — QUESTIONS

AF2-001
Question:
Solve: $4x + 9 = 33$
A) 5
B) 6
C) 7
D) 8

AF2-002
Question:
If $7n - 5 = 30$, what is n?
A) 4
B) 5
C) 6
D) 7

AF2-003
Question:
Simplify: $3(2x + 5) - 4x$
A) $2x + 15$
B) $6x + 15$
C) $10x + 5$
D) $2x + 5$

AF2-004
Question:
A supply company charges $12 per box plus a $20 delivery fee. Which expression represents the total cost for b boxes?
A) $12 + 20b$
B) $32b$
C) $12b + 20$
D) $20b - 12$

AF2-005
Question:
If $f(x) = 5x - 3$, what is $f(4)$?
A) 12
B) 17
C) 20
D) 23

AF2-006
Question:
Which ordered pair satisfies the equation $y = 2x + 1$?
A) (1, 4)
B) (2, 5)
C) (3, 6)
D) (4, 10)

AF2-007
Question:
Solve: 3(x - 2) = 21
A) 5
B) 7
C) 9
D) 11

AF2-008
Question:
A pattern begins: 6, 11, 16, 21, ... What is the next number?
A) 24
B) 25
C) 26
D) 27

AF2-009
Question:
Solve: x/6 = 8
A) 14
B) 24
C) 36
D) 48

AF2-010
Question:
A line has a slope of -2 and crosses the y-axis at 7. Which equation represents the line?
A) y = 7x - 2
B) y = -2x + 7
C) y = 2x + 7
D) y = -7x + 2

AF2-011
Question:
If 8 packages contain 96 bolts, how many bolts are in 5 packages at the same rate?
A) 40
B) 50
C) 60
D) 70

AF2-012
Question:
Solve: 12 - 3x = 0
A) -4
B) 0
C) 3
D) 4

AF2-013
Question:
Which expression is equivalent to 10a - 3a + 8?

A) 7a + 8
B) 13a + 8
C) 7a - 8
D) 10a + 5

AF2-014
Question:
A worker starts with 15 completed units and finishes 6 more units each hour. Which equation gives the total number of units U after h hours?
A) U = 15h + 6
B) U = 6h + 15
C) U = 21h
D) U = 15 - 6h

AF2-015
Question:
If y = 3x - 4, what is y when x = -2?
A) -10
B) -2
C) 2
D) 10

AF2-016
Question:
Solve: 6x + 4 = 2x + 28
A) 4
B) 5
C) 6
D) 8

AF2-017
Question:
Which value of x makes 4x - 1 greater than 15?
A) 3
B) 4
C) 5
D) 2

AF2-018
Question:
A table shows a linear relationship.

x: 0, 1, 2, 3
y: 5, 8, 11, 14

What is the rule?
A) y = 5x + 3
B) y = 3x + 5
C) y = 8x - 3
D) y = x + 5

AF2-019
Question:
Solve: $4(x + 3) = 2x + 20$
A) 2
B) 4
C) 6
D) 8

AF2-020
Question:
If $p = -5$ and $q = 2$, what is $3q - p$?
A) 1
B) 6
C) 11
D) -11

AF2-021
Question:
Which point is on the x-axis?
A) (0, 6)
B) (-4, 0)
C) (3, -3)
D) (5, 2)

AF2-022
Question:
A crew installs 18 brackets in 3 hours. At the same rate, how many brackets will the crew install in 7 hours?
A) 36
B) 42
C) 54
D) 63

AF2-023
Question:
Solve for t: $A = 5t + 10$
A) $t = A - 10$
B) $t = (A - 10)/5$
C) $t = 5(A - 10)$
D) $t = A/5 + 10$

AF2-024
Question:
What is the slope of a line that goes through (2, 3) and (6, 11)?
A) 1
B) 2
C) 3
D) 4

AF2-025
Question:

Simplify: $9x - 4 + 2x + 7$
A) $7x + 3$
B) $11x + 3$
C) $11x - 11$
D) $7x - 3$

AF2-026
Question:
If $g(x) = x^2 + 2x$, what is $g(3)$?
A) 9
B) 12
C) 15
D) 18

AF2-027
Question:
A sequence follows the rule “add 4, then multiply by 2.” If the first term is 5, what is the second term?
A) 9
B) 13
C) 18
D) 20

AF2-028
Question:
Solve: $2(x + 7) + 5 = 31$
A) 5
B) 6
C) 7
D) 8

AF2-029
Question:
A line crosses the y-axis at -3 and rises 4 units for every 1 unit to the right. Which equation represents the line?
A) $y = -3x + 4$
B) $y = 4x - 3$
C) $y = 3x - 4$
D) $y = -4x - 3$

AF2-030
Question:
If $4/7 = x/21$, what is x?
A) 8
B) 10
C) 12
D) 14

AF2-031
Question:
Solve: $-5x - 10 = -35$

A) -9
B) -5
C) 5
D) 9

AF2-032
Question:
A water tank contains 500 gallons and loses 25 gallons each hour. Which expression gives the amount of water left after h hours?
A) 500 + 25h
B) 500 - 25h
C) 25h - 500
D) 500 ÷ 25h

AF2-033
Question:
Which equation has x = 8 as its solution?
A) 2x + 3 = 21
B) 4x - 5 = 27
C) x/4 + 6 = 10
D) 3x + 2 = 25

PART B — ANSWER KEY + EXPLANATIONS

AF2-001 — Correct: B — Explanation:
Solve 4x + 9 = 33. Subtract 9 from both sides: 4x = 24. Divide by 4: x = 6. The most common mistake is subtracting 9 from 33 but then forgetting to divide by 4.

AF2-002 — Correct: B — Explanation:
Start with 7n - 5 = 30. Add 5 to both sides: 7n = 35. Divide by 7: n = 5. A common mistake is subtracting 5 instead of adding 5.

AF2-003 — Correct: A — Explanation:
Distribute first: 3(2x + 5) = 6x + 15. Then subtract 4x: 6x - 4x + 15 = 2x + 15. A common mistake is forgetting to multiply 5 by 3.

AF2-004 — Correct: C — Explanation:
The cost is $12 for each box, so that part is 12b. The delivery fee is a one-time charge of $20. The total cost is 12b + 20. A common mistake is putting the $20 fee with the variable as if it repeats for each box.

AF2-005 — Correct: B — Explanation:
Substitute 4 for x: f(4) = 5(4) - 3 = 20 - 3 = 17. A common mistake is adding 3 instead of subtracting it.

AF2-006 — Correct: B — Explanation:
Test the choices in y = 2x + 1. For (2, 5), 2(2) + 1 = 5, which matches y = 5. A common mistake is choosing a point where the result is close but not exact.

AF2-007 — Correct: C — Explanation:
Solve $3(x - 2) = 21$. Divide both sides by 3: $x - 2 = 7$. Add 2: $x = 9$. A common mistake is adding 2 before dividing, which does not undo the operations in the correct order.

AF2-008 — Correct: C — Explanation:
Each number increases by 5: 6, 11, 16, 21. Add 5 to 21: 26. A common mistake is assuming the pattern changes because the numbers are getting larger.

AF2-009 — Correct: D — Explanation:
$x/6 = 8$ means x divided by 6 equals 8. Multiply both sides by 6: $x = 48$. A common mistake is dividing 8 by 6 instead of multiplying.

AF2-010 — Correct: B — Explanation:
Slope-intercept form is $y = mx + b$. The slope is -2 and the y-intercept is 7, so the equation is $y = -2x + 7$. A common mistake is switching the slope and y-intercept.

AF2-011 — Correct: C — Explanation:
Find bolts per package: $96 \div 8 = 12$. For 5 packages: $12 \times 5 = 60$. A common mistake is multiplying 96 by 5 without first finding the unit rate.

AF2-012 — Correct: D — Explanation:
Solve $12 - 3x = 0$. Subtract 12 from both sides: $-3x = -12$. Divide by -3: $x = 4$. A common mistake is giving -4 because of the negative coefficient.

AF2-013 — Correct: A — Explanation:
Combine like terms: $10a - 3a = 7a$. The constant 8 remains unchanged. The expression becomes $7a + 8$. A common mistake is combining the 8 with the variable terms.

AF2-014 — Correct: B — Explanation:
The worker already has 15 units completed. Then 6 more units are completed each hour, so after h hours the total is $U = 6h + 15$. A common mistake is making 15 the hourly rate.

AF2-015 — Correct: A — Explanation:
Substitute $x = -2$ into $y = 3x - 4$: $y = 3(-2) - 4 = -6 - 4 = -10$. A common mistake is treating $3(-2)$ as positive 6.

AF2-016 — Correct: C — Explanation:
Solve $6x + 4 = 2x + 28$. Subtract $2x$: $4x + 4 = 28$. Subtract 4: $4x = 24$. Divide by 4: $x = 6$. A common mistake is subtracting 4 too early or from only one side.

AF2-017 — Correct: C — Explanation:
Solve $4x - 1 > 15$. Add 1: $4x > 16$. Divide by 4: $x > 4$. The only listed value greater than 4 is 5. A common mistake is choosing 4, but $4(4) - 1 = 15$, which is not greater than 15.

AF2-018 — Correct: B — Explanation:
The y-values increase by 3 each time x increases by 1, so the slope is 3. When $x = 0$, $y = 5$, so the y-intercept is 5. The rule is $y = 3x + 5$. A common mistake is using 5 as the slope because it is the first y-value.

AF2-019 — Correct: B — Explanation:
Distribute: $4x + 12 = 2x + 20$. Subtract $2x$: $2x + 12 = 20$. Subtract 12: $2x = 8$. Divide by 2: $x = 4$. A common mistake is forgetting to multiply 3 by 4.

AF2-020 — Correct: C — Explanation:
Substitute $p = -5$ and $q = 2$ into $3q - p$: $3(2) - (-5) = 6 + 5 = 11$. A common mistake is treating subtracting -5 as subtracting 5.

AF2-021 — Correct: B — Explanation:
A point on the x-axis always has $y = 0$. The point (-4, 0) has $y = 0$, so it is on the x-axis. A common mistake is choosing (0, 6), which is on the y-axis.

AF2-022 — Correct: B — Explanation:
18 brackets in 3 hours means $18 \div 3 = 6$ brackets per hour. In 7 hours: $6 \times 7 = 42$. A common mistake is adding 18 and 7 instead of using the rate.

AF2-023 — Correct: B — Explanation:
Start with $A = 5t + 10$. Subtract 10 from both sides: $A - 10 = 5t$. Divide by 5: $t = (A - 10)/5$. A common mistake is dividing A by 5 first and then subtracting 10.

AF2-024 — Correct: B — Explanation:
Slope equals change in y divided by change in x. From (2, 3) to (6, 11), y increases by 8 and x increases by 4. The slope is $8 \div 4 = 2$. A common mistake is using only the y-values and ignoring the change in x.

AF2-025 — Correct: B — Explanation:
Combine like terms: $9x + 2x = 11x$, and $-4 + 7 = 3$. The simplified expression is $11x + 3$. A common mistake is subtracting $2x$ because the previous constant is negative.

AF2-026 — Correct: C — Explanation:
Substitute 3 for x: $g(3) = 3^2 + 2(3) = 9 + 6 = 15$. A common mistake is calculating 3^2 as 6 instead of 9.

AF2-027 — Correct: C — Explanation:
Start with 5. Add 4: 9. Then multiply by 2: 18. A common mistake is multiplying first and then adding 4, which gives 14.

AF2-028 — Correct: B — Explanation:
Solve $2(x + 7) + 5 = 31$. Distribute: $2x + 14 + 5 = 31$. Combine: $2x + 19 = 31$. Subtract 19: $2x = 12$. Divide by 2: $x = 6$. A common mistake is forgetting to add 14 and 5 before solving.

AF2-029 — Correct: B — Explanation:
The line rises 4 units for every 1 unit to the right, so the slope is 4. It crosses the y-axis at -3, so the y-intercept is -3. The equation is $y = 4x - 3$. A common mistake is switching the slope and intercept.

AF2-030 — Correct: C — Explanation:
Use the proportion $4/7 = x/21$. Since $7 \times 3 = 21$, multiply 4 by 3: $x = 12$. A common mistake is adding 14 to the numerator because 7 becomes 21.

AF2-031 — Correct: C — Explanation:
Solve $-5x - 10 = -35$. Add 10 to both sides: $-5x = -25$. Divide by -5: $x = 5$. A common mistake is giving -5 because both sides contain negative numbers.

AF2-032 — Correct: B — Explanation:
The tank starts with 500 gallons and loses 25 gallons each hour. After h hours, the amount left is $500 - 25h$. A common mistake is adding 25h even though the water amount is decreasing.

AF2-033 — Correct: B — Explanation:
Test $x = 8$ in each equation. In choice B, $4(8) - 5 = 32 - 5 = 27$, so the equation is true. A common mistake is choosing a nearby equation without checking both sides exactly.